

JUNAID KHAN

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PROFESSIONAL SUMMARY

Software Development Engineer specializing in Java backend microservices, relational database architecture, and AI-driven applications. Experienced in developing scalable RESTful APIs with **Spring Boot** and **Spring Security 6**, managing complex schemas using **PostgreSQL/JPA** and vector stores like **pgvector**, and integrating third-party systems like **Razorpay** and **OAuth 2.0**. Proficient in full-stack development with **React** and incorporating **LangChain / RAG** pipelines for context-aware AI capabilities.

TECHNICAL SKILLS

Languages:	Java, Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
Backend Engineering:	Spring Boot, Spring Security 6, Spring Data JPA, Hibernate, REST APIs, Gradle
Databases & Auth:	PostgreSQL, pgvector, MongoDB, JWT, OAuth 2.0, Google OAuth, RBAC, bcrypt
AI / ML Integration:	LangChain, RAG (Retrieval-Augmented Generation), OpenAI Whisper, Groq / Gemini APIs
Frontend & Tools:	React.js, Redux, Tailwind CSS, Git, GitHub, Docker, FFmpeg, Linux

KEY PROJECTS

LearnVault — AI-Powered E-Learning Platform | *Java, Spring Boot, React, LangChain, PostgreSQL, Razorpay* [GitHub Repository](#) | [Live Deployment](#)

- Architected an end-to-end e-learning platform with secure course delivery, role-based user management (Student/Instructor), and dynamic content management.
- Implemented **Spring Security 6** with **JWT** and **OAuth 2.0** for stateless authentication, endpoint protection, and granular Role-Based Access Control (RBAC).
- Designed optimized relational database schemas using **Spring Data JPA** and **PostgreSQL**, reducing API response latencies via effective indexing and lazy loading.
- Integrated **Razorpay Payment Gateway** with webhook event listeners for secure checkout transactions and automated enrollment workflows.
- Engineered an embedded **RAG (Retrieval-Augmented Generation)** assistant using **LangChain** and **pgvector** vector embeddings, enabling interactive, context-aware student query resolution.

VideoClipper — AI-Powered Short-Form Video Generator | *Python, Flask, FFmpeg, Whisper* [GitHub Repository](#) | [Live Deployment](#)

- Engineered an automated video processing pipeline extracting high-engagement clips from long-form content for YouTube Shorts, Instagram Reels, and TikTok.
- Developed a multi-dimensional text and keyword analysis scoring engine evaluating key phrases, emotional hooks, and narrative arcs to identify viral segment timestamps.
- Integrated **OpenAI Whisper** for local audio transcription with fallback to **Gemini/Groq APIs**, generating word-level synced **SRT subtitles** burned via **FFmpeg**.
- Implemented dynamic aspect ratio transformation (16:9 to 9:16) with smart background blur and centered subject tracking.

PromptKit — LLM Text Preprocessing Toolkit | *Python, PyPI Package* [GitHub Repository](#) | [PyPI Package Link](#)

- Developed a lightweight Python library for sanitizing, stripping noise, and standardizing prompt inputs prior to LLM tokenization, reducing token consumption and enhancing output reliability.

ACHIEVEMENTS & CONTRIBUTIONS

- **LeetCode:** Solved data structures & algorithms problems across trees, dynamic programming, recursion, backtracking, and system design fundamentals.
- **Open Source / GitHub:** Earned *Pull Shark x2* badges; active contributor to modular backend and utility open-source tools.

EDUCATION

Master of Computer Applications (MCA) — Sikkim Manipal University

New Delhi, India